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How can we use artificial intelligence for stock recommendation and risk management? A proposed decision support system

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ABSTRACT

Background: Decision-making in the stock market is convoluted as it requires significant trading experience and knowledge. Faced with a huge range of stocks, investors in the stock market may be overwhelmed and daunted by the number of choices available on their plates. Financial institutions also face the problem of recommending stocks that cater to the investors' trading strategies.

Objective: In this paper, we develop and explore three different approaches that can be used to build a stock recommender system that takes into account the investors' needs and interests. Furthermore, to ensure the profitability of the recommended portfolios and to support the investors in making risk-informed decisions, we evaluate their Expected Returns and Value-at-Risks.

Methods & Findings: Hierarchical clustering was performed to better understand groups of traders that are similar in their needs and preferences and thus computational efficiency was improved. Three stock recommender systems, the K-Nearest Neighbour (kNN), Singular Value Decomposition (SVD) and Association Rule Mining (ARM) was explored and evaluated. Our short term, medium term and long term portfolios average rate of return was 4.15%, 10.24% and 23.17% respectively.

Conclusion: Our study demonstrates promising recommendation results that not only catered to the user's profile but also contributed to the portfolio profitabilities with minimal financial loss. Stock recommendations were beneficial to users when their preferences were considered.

1. Introduction

Riding on the crest of technology- social media and the World Wide Web have become prevalent. The expeditious and extensive adoption of these technologies has changed how we interact with one another and make decisions in our daily lives. While this increase in online traffic has allowed people in different corners of the world to connect seamlessly, the problem of information overload has transpired (Edmunds & Morris, 2000). This excessive increase in available information is overwhelming and hinders decisiveness. Therefore, there is now a need to filter and provide relevant information to improve people's online experience, escalating the ever-demand for recommender systems.

Recommender systems are information filtering systems that aim to alleviate the problem of information overload by extracting valuable patterns and information from a large amount of data and then providing item suggestions that may be of best interest to the user (Ricci et al., 2011). With the growth of customer-driven marketing, business companies aim to maximize customer satisfaction by increasing the demand for personalized recommender systems that learn users' tastes and preferences and produce outcomes that correlate to these interests, easing the decision-making process (Isinkaye et al., 2015). Recommendation systems were developed in many industries such as e-commerce, banking, retail, and media, showing their success in improving customer satisfaction and business value (Lee & Hosanagar, 2017; Amatriain & Basilico, 2015). For example, the popular video streaming platform Youtube reports that users spend 70% of their time on the platform watching videos recommended by their algorithm (Ye, 2020).

Recommender systems are of great interest in the stock market to help investors with their investment decisions. Stock trading is a form of investment where investors assemble to buy and sell company shares in the stock market. The market is highly volatile and heavily dependent on economic events. Significant trading experience and being up to date on financial news are crucial for successful trading. For this very reason, investing in the stock market to achieve financial gains is a challenging task. Moreover, with thousands of stocks available in the stock market, investors may be overwhelmed on which stocks to invest in. Floundering in a maze of information further affects the traders decision-making capabilities, increasing their risk of investing in the wrong stocks. In

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addition, investors, especially those beginners, who lack the required knowledge and experience to trade successfully in the stock market, often rely on the media to empower them with information about which stocks seem most promising (Fang & Peress, 2011). Consequently, recent literature suggests that investors invest in "attention-grabbing" stocks covered in the media (Barber & Odean, 2008). However, such recommendations may entail high risks, especially if followed blindly, resulting in massive financial losses.

While it is important to provide profitable stock recommendations, the users' preferences and satisfactions should also be considered to be an effective stock recommender system. For example, recommending popular stocks may not be helpful as these may already be known to the user. Similarly, having small-cap stocks recommended to low-risk investors would be unsuitable and frustrating as it does not align with their interests. Hence, it is essential to have a flexible recommender system that not only minimises the users' financial losses but is also able to provide personalized recommendations that aid in their decision-making and enhance their overall experience.

Therefore, this study proposes a personalized recommender system that offers value to laypersons investing in stocks. We first segment our traders using hierarchical clustering and then apply collaborative filtering to recommend stocks to our traders. Three collaborative filtering models were built to provide personalized stock recommendations to the users based on their past trading history. The results were evaluated based on predictive and classification accuracy metrics namely Mean Absolute Error, Root Mean Squared Error and Mean Average Precision at 10 in order to find the best model that is able to predict the users' ratings well and provide relevant recommendations that suit their taste and preferences. The recommendations from the best model were then evaluated based on their expected rate of return and Monte Carlo simulation for value-at-risk to show that they are profitable and will not result in massive financial losses to the users.

All in all, this paper aims to show how a stock recommender system can be used as a decision support system that can help traders make trading decisions by analysing and suggesting similar stocks that they may be interested in buying. With the vast number of stocks available in the stock market alongside the number of online resources at the traders' fingertips that can "help" to identify which stocks may be beneficial to them and would suit their trading strategies, having a recommender system that can provide personalized stock suggestions would alleviate the problem of information overload and ease the decision-making process for these traders, saving their time and energy. Since our recommender system provides stock suggestions that are catered to the trader's interests, this personalization further strengthens the effectiveness of our model in aiding the trader's stock choices. Furthermore, through simulations of the possible risks and returns that these traders can experience, our model can present the best possible trading decisions to make and they can be rest assured that our model's stock suggestions would result in financial gains for them.

While Liu (2017) also built a recommender system for retail trading using the same data, different techniques were used in our study. We used hierarchical clustering to segment our traders and the collaborative filtering technique to recommend stocks to our traders. Furthermore, our study acknowledges that different traders have different investment styles and hence aims to cater to the different trading behaviors of these users by segmenting them into whether they are short-term, medium-term, and long-term traders. Lastly, to ensure that the recommended stocks do not bear massive risks, we also evaluated the recommended portfolios' Value-at-Risk, thereby supporting the investors in making risk-informed decisions and enabling them to control their market exposure.

Our paper is structured as follows: In Section 2, we briefly summarize the literature review. Section 3 introduces the dataset, data cleaning process, and covers the methods used to build a recommender system. Section 4 highlights and discusses the results, and Section 5, evaluates the expected returns and the potential risks derived from the recommended portfolios. Section 6 discusses the theoretical and practical implications, limitations, and possible future works of our study. Section 7 provides the conclusion of our paper.

2. Related works

To ensure that the businesses realize the real value of Big Data Analytics, the energy and resources must focus on collecting big data and generating insights, but it is also crucial to ensure that these insights are implemented on a real-time basis (Amit Kumar, et.al, 2021). The research articles analyzed in the current study (Palmieri et al., 2016; Sbai & Krichen, 2020; Ghosh & Gosavi, 2017) have reported through various mechanisms on how Big Data Analytics can be converted into actionable insights that the business can harness the value of Big Data Analytics from their investments. Kushwaha et. al (2021) also provides a review of the increasing applications of Big Data Analytics to improve management practices, highlighting the growing significance of text mining and its power to improve business performance. Chand and Zhang (2022) showed how the combination of Big Data Analytics with machine learning models can not only improve the decision-making process of the business but also cut down its resource spending. Rawat et al. (2021) applied Big Data Analytics through machine learning and the visualization of data to evaluate the claims made in the insurance sector. Dahiya and Dalal (2022) proposed an improvement in mining Big Data that is often inefficient due to memory and processing capability issues by using a parallel framework to extract large data, thereby allowing for insights to be garnered in real-time. In our study, we would also like to demonstrate how the application of Big Data Analytics can be used to recommend stocks to traders based on their preferences. Further, the research on Artificial Intelligence (AI) applications in the business-to-customer (B2C) market has been seen extensively. For example, the application of augmented reality and mixed reality to provide a realistic and futuristic shopping experience to the customer (Hoyer, Kroschke, Schmitt, Kraume, & Shankar, 2020; Pillai, Sivathanu, & Dwivedi, 2020), the influence of virtual reality and artificial intelligence on consumer learning (Suh & Lee Suh & Lee, 2005). Moreover, Sharma et. al (2021) studied how a personalized recommender system can be built based on semantic measures and showed that this can transcend traditional recommender systems, enhancing a user's overall experience. In the same way, our stock recommender system approach aims to generate a customer-specific experience by recommending stocks to our users based on their stock preferences.

There has been increasing research on existing recommender systems applied in the financial sector. We, therefore, will first review the collaborative filtering and content-based approaches used to build a stock recommender system.

Collaborative filtering (CF) is one of the most popular techniques used to build recommender systems. It works on the assumption that users with similar preferences in the past are more likely to have similar interests in the future. There are two categories: User-based and Item-based. User-based CF finds users with similar consumption patterns and recommends items that these similar users find interesting. User-based CF performance decreases when we have sparse datasets (Boström & Filipsson, 2017). On the other hand, item-based CF recommends items similar to what the user has liked in the past. Since the properties of items remain more constant compared to users' evolving tastes and preferences, item-based CF tends to be more scalable than user-based CF (Boström & Filipsson, 2017).

Sayyed et al. (2013) presented a preliminary investigation on the implementation of CF methods on the stock market but had not validated CF methods on stock data. Vismayaa et al. (2020) compared the performance of single classifiers against ensemble classifier-based recommender systems on stocks from the Bombay Stock Exchange based on classification accuracy and economic measures. This study

showed that ensemble classifier-based recommender systems surpass single classifiers in predicting stock price movements. Prieto-Torres and Galpin (2020) used a CF model to build a virtual wallet recommender system for a FinTech company to enhance the user experience on their mobile application. However, this method worked best using a non-personalized approach where the top-N most popular stocks were recommended to each user. Anwar, et.al, (2019) proposed a collaborative filtering based movie recommendation using rule mining to satisfy individual's requirement. Their paper also provided a comparative analysis of the several types of similarity measures namely Cosine, Correlation, Euclidean, Jaccard and Manhattan. The correlation similarity measure yielded the best precision, recall and F1-score. Further, Anwar, et.al, (2021) introduced a new approach that uses CF and Singular Value Decomposition (SVD)++ for implementing a recommendation system. Their proposed approach gave a smaller error rate when cross-validation (CV= {5, 10, 15}) was performed. Their proposed approach also alleviated the sparsity and cold-start problems. Kirubahari, R, et.al (2021) proposed a weighted parallel deep hybrid collaborative filtering approach based on Singular Value Decomposition (SVD) and Restricted Boltzmann Machine (RBM). Their results indicated better prediction compared to other techniques in terms of accuracy. Cui et. al (2019) also explored the traditional CF approach as well as SVD to ease fund companies' decision-making process in identifying which listed company to invest in. In their study, they tackled the issue of sparsity for user-based CF by taking into account the number of items that are similarly scored by the users and improved traditional item-based CF by inputting the missing ratings of the users through an iterative process instead of just using the mean ratings. Their study showed that the improved item-based CF outperforms the other two methods. As CF typically suffers from the cold-start problem, Rajendran and Sundarraj (2021) proposed a hybrid CF model that incorporated the browsing history of the users alongside the usual ratings as input and showed that when there is insufficient information about a user, the accuracy improves when their enhanced CF model is used. Lastly, Rukiya et. al (2019) proposes an improvement to the traditional CF approach by first clustering based on the popularity of the items and segmenting the users based on their loyalty to the business to provide a more personalized recommendation while ensuring that items that are already known to the user are not recommended. This allows e-commerce platforms' targeted marketing approaches to be more effective, garnering greater user loyalty.

Content-based filtering (CBF) is another technique used to build recommender systems. It uses the information about the items obtained from the user's profile to recommend similar items to what the user has liked in the past. Unlike CF, CBF does not take into account other users' information. Yoo et al. (2003) proposed a personalized filtering system that only considers the trading information that a user deems relevant. Personalized recommendations based on the Moving Average Convergence Divergence (MACD) advise traders whether to buy or sell a stock by examining the difference between two moving averages. Chalidabhongse and Kaensar (2006) proposed an adaptive user model that performs stochastic technical analysis to forecast stock returns while considering explicit preferences and user interactions with the system for personalized recommendations. Rutkowski (2021) built a Neuro-Fuzzy recommender system to recommend a stock based on similarity from a user's past transaction if the stock passed the fuzzy set of rules generated.

3. Methodology

In this section, we present the dataset and the cleaning and preprocessing steps that we have done. Thereafter, we elaborate on the proposed methods that can be used to build a stock recommender system. Fig. 1 shows an overview of this section.

3.1. Data collection, cleaning and processing

The dataset, more commonly known as the LDB Data, comes from a large discount brokerage firm consisting of the trading records of 78,000 households from January 1991 to December 1996 that Barber and Odean (2000) used in their study. Investments that are not in common stocks, such as mutual funds, warrants, and options, were filtered out since the primary focus of our study will be on equity investments. Our study retained 62,963 out of the 78,000 household investments. Furthermore, each household could have more than one account due to different accounts for each household member. For instance, a parent can create a custodial account for their child, or some could also open a retirement account (Barber & Odean, 2001). Therefore, in this study, we will focus our analysis and apply our methodologies for each account.

While we acknowledge that the dataset that is currently being used is outdated, due to our hopes of building a recommender system based on the users' tastes and preferences, obtaining users' transactional data was of utmost importance. However, due to privacy issues, having access to such data is a challenge and this was the best that we could find. Nevertheless, our study aims to show that with the transactional data of the traders, we are able to build a customer-centric recommender system that is able to suggest stocks that align with the trader's interests.

We omit trading records with negative commissions as this may indicate that the broker has reversed the transaction.

Different trading strategies and preferences exist for short-term, medium-term, and long-term traders. Long-term traders are generally less open to investing in small-cap stocks, as these stocks tend to carry more risk and tend to invest in large-cap stocks that entail less risk. Instead, we first need to assess the trading strategies our users follow so that we can cater more to their needs and interests. Therefore, in this study, if a trader holds a stock for less than three months on average, we define him/her to be a short-term trader. We consider him/her a medium-term trader if he/she holds a stock for more than or equal to 3 months but less than one year on average and a long-term trader otherwise. Table 1 below shows the trading behaviors of each type of trader.

Table 1 shows that 60% of the app users are medium-term traders and long-term traders. These traders had greater profits than short-term traders. Furthermore, the mean and median profits earned by short-term traders differed more than the other two segments, indicating that short-term traders experience more variation in their gains and losses and are more vulnerable in terms of their risk exposure. It further strengthens the need to provide a recommender system for them that carry lesser risks. Upon further exploration of the data, it seems that many users have too few trading episodes to indicate their tastes and preferences, and may not be the true users of our Stock app, hence we omit those that have made less than five trades. Furthermore, as we also want to provide profitable recommendations to the users, assuming that the users who earned profits from their investments traded in stocks that have good returns, we should build our recommender system using only these groups of users. Hence, we focus on those traders who have earned a positive average rate of return.

In addition, deeper analysis of the user profiles of those that have been excluded showed that the stocks that these users are investing in vary significantly from the stocks that those who have earned a positive average rate of return are putting their money into. This tells us that these users may not be aware of the more favourable stocks that have greater potential for higher returns. This further supports the need to exclude them when building our recommender system to avoid stocks with similarly low rates of returns from being recommended. For this group of users, it might be better to take a non-personalized approach instead – where the most popular stocks that the users who have benefitted from their investments are investing in are recommended, with the hope that they will get better rates of return.

Since the tastes and preferences of the users are constantly evolving, to keep in line with the current economic conditions, we assume that

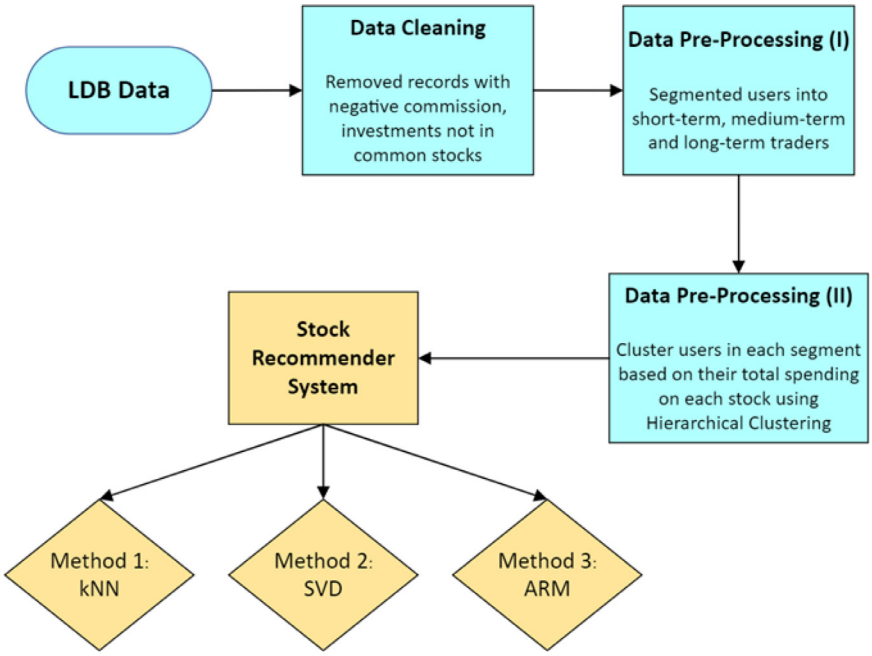

Fig. 1. Methodology flowchart

Table 1
Comparison between Short-Term, Medium-Term and Long-Term Traders

	Short-Term Traders	Medium-Term Traders	Long-Term Traders
No. of Traders	6,506 (11.5%)	35,273 (62.6%)	14,602 (25.9%)
Mean Profit (± S.D.)	$4,853 (± $4,866)	$7,060 (± $5,862)	$6,956 (± $3,947)
Median Profit	$2,797	$6,574	$5,388
Number of Traders that earned a profit	4,128 (63.45%)	23,339 (66.17%)	9,256 (63.38%)
Mean Holding Days (± S.D.)	34.5 (± 17.4)	195 (± 52.7)	588 (± 211.5)
Median Holding Days	35	188	521
Mean Latency between 1st and 2nd Trade (± S.D.)	39 days (± 25 days)	60 days (± 49 days)	93 days (± 88 days)
Median Latency between 1st and 2nd Trade	31 days	56 days	120 days

a short-term, medium-term, and long-term trader's trading preferences will not change over 6, 12, and 18 months respectively. We grouped traders into short, medium, and long-term traders based on their current interests and previous trading activities and provided relevant recommendations. Therefore, in this study, we will be focusing on these respective timeframes to build and evaluate our recommender systems for each segment.

3.2. Data clustering

Our Stock app has a high number of users and stocks and, therefore, our dataset currently suffers from high sparsity and poor scalability. Much of the information about the preferences of the app user is not present since it is hard to achieve a situation where a user has bought every single stock available in the stock market (de Aguiar et al., 2020). Furthermore, depending on the economic conditions, the amount and type of stock bought vary quite vastly, further worsening the degree of sparsity of the data (Hayashi, & Takahashi, 2021). Since the computations grow exponentially as the number of users and items in our data increases, the computations required to run our model will be costly, affecting the scalability of our recommender system and worsening the system's performance and the users' experience. To solve this problem, we propose clustering the users based on their total spending on each stock before building a recommender system for each of these clusters. Agglomerative hierarchical clustering is a popular cluster method that aims to achieve maximum within-cluster similarity and minimum between-cluster similarity by presenting the data in a tree structure called dendrograms. Agglomerative hierarchical clustering performs a bottom-up approach where each pair of clusters is combined in each iteration until one big, single cluster is formed (Johnson, 1967). While different types of linkages can be applied to measure the pairwise distances between clusters, in this study, we use Ward's Method, which chooses the pair of clusters to merge based on which results in the minimum total-within cluster variance (Ward, 1963). We visualize the hierarchical clusters using a dendrogram, as seen in Figs. 2–4.

We calculate the cophenetic correlation coefficient to evaluate the goodness of fit or the quality of the clusters generated. The cophenetic correlation coefficient ranges from –1 to 1, with 1 indicating an excellent fit, and is calculated by equation 1 below (Saraçli et al., 2013):

$$c = \frac{\sum_{i<j}(x(i,j) - x)(t(i,j) - t)}{\sqrt{\left[\sum_{i<j}(x(i,j) - x)\right]^2 \left[\sum_{i<j}(t(i,j) - t)\right]^2}} \quad (1)$$

where x(i,j) is the Euclidean distance between the ith and j^{th} observation, x is the average of x(i,j), t(i,j) is the dendrogrammatic distance between the ith and j^{th} observation and t is the average of t(i,j).

To determine how many clusters to use for the short-term traders, medium-term traders, and long-term traders, we calculated the cophenetic correlation coefficient with k = 3, 4, and 5.

From Table 2, k = 3 gives us the best clustering results for the short-term segment. While k = 4 gives us the best clustering results for the medium-term and long-term segments. Hence, we built a recommender system for each of these clusters in their respective segments.

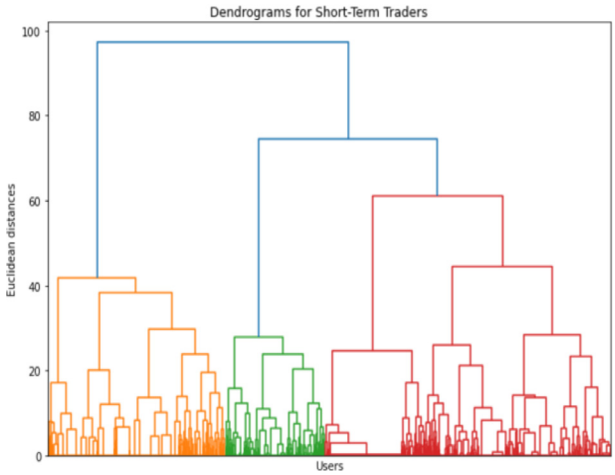

Fig. 2. Dendrograms for Short-Term Traders

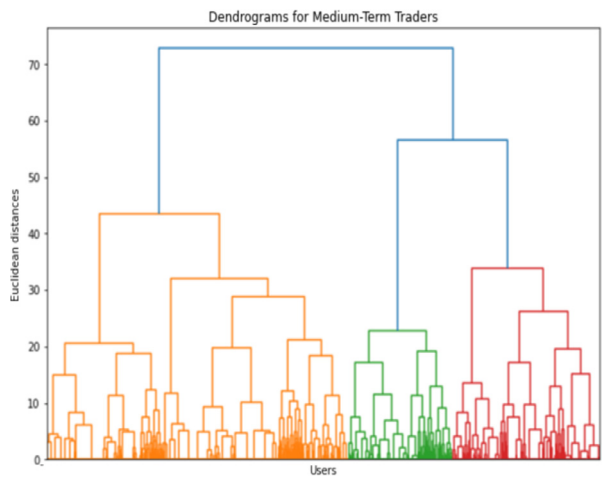

Fig. 3. Dendrograms for Medium-Term Traders

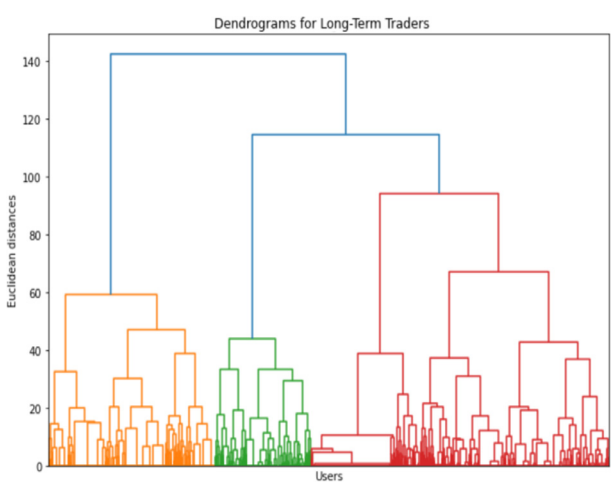

Fig. 4. Dendrograms for Long-Term Traders

Table 2
The goodness of fit results for hierarchical clustering

k	Cophenetic Correlation Coefficient		
	Short-Term	Medium-Term	Long-Term
3	0.82	0.79	0.81
4	0.79	0.83	0.85
5	0.74	0.81	0.77

3.3. k-Nearest Neighbor

Another challenge we face with our dataset is the lack of explicit ratings from the users to indicate their preference over the stocks that they invest in, which we would require when building our recommender system. Therefore, to tackle this challenge, we assume that the greater the number of stocks bought by an app user, the higher their user preference for that stock. With this assumption, we estimate every user i' explicit rating for stock j using equation 2 below (Liu, 2017):

$$r_{ij} = 1 + 4\frac{\sum_k Q_{ij}^{(k)}}{\sum_{j,k} Q_{ij}^{(k)}} \tag{2}$$

where the numerator $\sum_k Q_{ij}^{(k)}$ represents the total quantity of stock j bought by user i in his kth transaction and the denominator $\sum_{j,k} Q_{ij}^{(k)}$ is the total quantity of all the stocks that user i has bought.

Now that we have the explicit ratings, we can conduct the usual recommender system algorithms.

The k-Nearest Neighbour (kNN) approach is a popular technique for building a collaborative filtering (CF) system. The kNN approach is a non-parametric, lazy learning algorithm that trains the dataset whenever a recommendation is needed (Guo et al., 2003). The kNN approach aims to find the k-most similar users or items using the Pearson or cosine similarity measures. In this study, we perform an item-based kNN approach and use a cosine similarity which is computed by equation 3 below:

$$\text{Cosine Similarity}: \; sim(i,j) = \frac{\sum_{u=1}^{m} r_{ui}\, r_{uj}}{\sqrt{\sum_{u=1}^{m} r_{ui}^2 \sum_{u=1}^{m} r_{uj}^2}} \tag{3}$$

where r_{xy} is the rating provided by user x on item y.

How well the model performs depends on finding the most optimal value of k – too large a value would result in being too computationally expensive and underfitting, thus leading to poor prediction performance. Too small a value for 'k' leads to overfitting and poor performance on unseen data. Hence to determine the optimal value of k, we plot the error rates for a given range of k and choose the optimum k to be the one with the least error rate to be the optimum (Zhang, 2016). Furthermore, the value of k chosen should be odd to avoid situations that involve a tie.

After specifying k, consequently, we use their weighted average to calculate the predicted rating for target user i on item j using equation 4 below:

$$\hat{r}_{ij} = \overline{r_i} + \frac{\sum_k sim(j,k)\left(r_{ik} - \overline{r_k}\right)}{\sum_k |sim(i,k)|} \tag{4}$$

where $\overline{r_i}$ is the average rating of user i.

We next provide recommendations based on the top N items with the highest predicted ratings for each user.

3.4. Singular value decomposition

Apart from kNN, another CF approach that has gained popularity in the recent years is Singular Value Decomposition (SVD). It is a dimension reduction, matrix factorization technique widely used as a CF technique in the Netflix Prize competition with the hopes of improving Netflix's current recommender system and was shown to be outperforming traditional CF methods by about 10% (Gower, 2014). As such, in our study, we also explore the performance of SVD as a stock recommender system and evaluate its performance against other CF models.

SVD aims to extract latent variables to decompose the user-item interaction matrix into vectors of user and item factors separately. The predicted rating that user i will give item j is then calculated as the dot product of the corresponding user and item feature vectors (Zhao et al., 2007), as shown in equation 5.

$$\hat{r}_{ij} = \mu + b_i + b_j + q_j^T p_i \tag{5}$$

where μ is the users' average rating, b_i and b_j are the bias from user i and item j respectively, q_j represents the feature vector for item j and p_i represents the feature vector for user i.

To estimate the unknowns, we solve the following optimisation problem using equation 6 below:

$$min \sum_{r_{ij} \epsilon R_{train}} (r_{ij} - \hat{r}_{ij})^2 + \lambda \left(b_i^2 + b_j^2 + \left|\left|q_j\right|\right|^2 + \left|\left|p_i\right|\right|^2\right) \tag{6}$$

where R_{train} is the matrix obtained from the training data and λ is the regularisation term to prevent overfitting.

To solve the minimisation problem, we use the stochastic gradient descent (SGD) by recursively solving equations 7, 8, and 9 below:

$$e_{ij} = r_{ij} - \hat{r}_{ij} \tag{7}$$

$$q_j \leftarrow q_j + \gamma \left(e_{ij} \cdot p_u - \lambda q_j\right) \tag{8}$$

$$p_u \leftarrow p_u + \gamma \left(e_{ij} \cdot q_j - \lambda p_u\right) \tag{9}$$

After obtaining q_j and p_i, we can next calculate $\hat{r}_{ij}$ and recommend the top N items with the highest predicted ratings for each user.

3.5. Association rule mining

Association Rule Mining (ARM) is a valuable data mining technique for unveiling hidden relations among items by using the users' transaction history (Hikmawati et al., 2020) and has been widely used in domains such as market basket analysis and customer analytics, especially when explicit ratings are not available.

An association rule X Y where X and Y, are item sets indicating that a user buying X is also likely to buy Y in the same transaction. It is typically measured based on its Support – which measures how often the rule occurs in all the transactions (See equation 10 below) and Confidence -– which measures the conditional probability of a user buying item Y given the user bought item X (See equation 11 below).

$$support(X \rightarrow Y) = \frac{number\ of\ transactions\ containing\ X\ and\ Y}{total\ number\ of\ transactions} \tag{10}$$

$$confidence(X \rightarrow Y) = \frac{number\ of\ transactions\ containing\ X\ and\ Y}{number\ of\ transactions\ containing\ X} \tag{11}$$

We originally did not have the users' explicit ratings and therefore had to estimate them as seen in equation 4 in order to build our SVD and kNN models. Unlike the usual Collaborative Filtering methods, however, Associate Rule Mining (ARM) uses implicit feedback where the tastes and preferences of the users are inferred from their previous transactions. Previous studies have shown that implicit feedback models can produce satisfying results (Hu et. al, 2008; Najafabadi et. al, 2017; Schoinas et. al, 2019). Hence, in this study, we also compare and evaluate the performance of CF methods using explicit feedback as opposed to implicit feedback.

ARM models the user's transaction as a binary rating matrix, with a '1′ indicating that a user has previously bought that item and a '0′ otherwise. To provide recommendations to a user u, given the set of items T_u that the user likes and the set of all association rules R, the following steps are taken (Hahsler, 2021):

- Specify the minimum support and confidence values as mentioned above.
- Find all relevant rules X Y for which X $\subseteq T_u$ in R.
- Recommend the top N items from the consequents (Y) of these rules in descending confidence values.

4. Results

4.1. Evaluation metrics

Since the quality of the recommendations provided can significantly affect the users' experience and the business value, choosing the final model using well-established evaluation metrics that are beneficial and meet our objectives. In this study, we use the following evaluation metrics to compare the performance of our algorithms.

The first metric we use is the mean absolute error (MAE) which is given by equation 12 below:

$$MAE = \frac{1}{n} \sum | r_{ij} - \hat{r}_{ij} | \tag{12}$$

where n is the total number of users in the dataset, r_{ij} is the true rating and $\hat{r}_{ij}$ is the predicted rating.

We also look into the root mean squared error (RMSE) that is defined by equation 13 below:

$$RMSE = \sqrt{\frac{1}{n} \sum (r_{ij} - \hat{r}_{ij})^2} \tag{13}$$

The Mean Absolute Error and Root Mean Square Error are explicit ratings and cannot be applied to evaluate recommender systems that use implicit ratings. We, therefore, cannot directly compare the performance of the implicit and explicit recommender systems. In order to have a more comparable evaluation metric, we use the mean average precision at 'k' (mAP@k) metric. This metric aims to measure the relevance of the recommended k items while considering the order of the items in the list and disregards the exact rating made by the user, allowing us to evaluate models that use either implicit or explicit data as input.

To calculate mAP@k, we first need to calculate the average precision (AP) at a threshold of k items using equation 14 below (Sawtelle, 2016).

$$AP\ @\ \mathrm{k} = \frac{1}{m} \sum_{i=1}^{k} Precision\ (i) \cdot relevant(i) \tag{14}$$

where Precision(i) measures the proportion of recommended items that are relevant in the top i set, relevant(i) is an indicator function with 1 being item i is relevant and 0 otherwise and m is the total number of items that are relevant to the user.

Then mAP@k is simply the average of AP@k among all users in the dataset and is given by equation 15 below:

$$mAP\ @\ k = \frac{1}{N} \sum_{i=1}^{N} AP@k_i \tag{15}$$

4.2. Evaluation of results

To validate our recommender systems, for the short-term traders, we set the training data to be the records based on their past four months of data and test it on the next two months of their data. For the medium-term traders, we train the system using nine months of their past data and apply it to predict which stocks each user would buy in the following three months. Lastly, for the long-term traders, we train our recommender system based on their last twelve months of trading data and test it on their next six months of trading activities.

K-fold cross-validation (Refaeilzadeh et al., 2009) was performed using the training set, with K = 10. We first partitioned the dataset into K equal subsets, and of these K subsets, we retained one subset as a validation set, and the remaining K–1 subsets were used as the training set. The above process was repeated K times, using a different validation subset for K = 1, 2, 3, ..., 10. We then evaluated how our recommender system performed on a new unseen dataset, the test set. Table 3 shows the aggregated results from the respective clusters based on each segment.

Table 3 shows that the recommender system performs better when explicit feedback is used instead of implicit feedback. According to (Hu et al., 2008), implicit feedback is a one-class problem that only

Table 3
Results summary for the different recommender systems for each segment

Segment		ARM	*k*NN	SVD
Short-Term	mAP@10 (%)	31.22	75.91	72.80
	RMSE	-	0.70	1.12
	MAE	-	0.57	0.78
Medium-Term	mAP@10 (%)	36.24	80.69	74.98
	RMSE	-	0.45	0.61
	MAE	-	0.38	0.60
Long-Term	mAP@10 (%)	39.59	83.87	76.57
	RMSE	-	0.49	0.68
	MAE	-	0.38	0.55

Table 4
Expected rate of return

Segment	Average Expected RoR
Short Term (30 days)	4.15%
Medium Term (200 days)	10.24%
Long Term (580 days)	23.17%

considers positive feedback and has noise. We note that our dataset also has high sparsity, which further worsens the performance of our implicit feedback recommender system and thus does not perform as well compared to the systems using explicit ratings. We can also see that the kNN approach performs better for all metrics when compared to the SVD approach. It could be due to the extraction of the latent factors, which may have ignored important information. Therefore, we conclude that using kNN to build a recommender system gives us the best results and use this as our final model when evaluating the profitability of the recommendations in the next section.

5. Portfolio optimisation

While MAE, RMSE, and mAP@10 help us assess the relevance and quality of the recommendations, users would want to know the potential gains and losses they may experience. We, therefore, want to evaluate next the profitability and the financial risks that our recommendations may bring to our users. Therefore, we assess the Expected Rate of Return and Value-at-Risk of these recommended stocks.

5.1. Expected rate of return

To assess the profitability of the recommended stocks, we calculate the expected rate of return (RoR) for a portfolio, $\bar{R}_p$, over a specified period as (Brenyah, 2017) using equation 16 below:

$$\bar{R}_p = \sum_{j=1}^{m} W_j R_j \ x \ 100\% \tag{16}$$

where W_j is the weight, or the proportion, of stock *j*, R_j is the expected (mean) return of stock *j* and m is the total number of stocks in the portfolio.

For our study, since we do not know the actual values, we generate random weights for the stocks and calculate the average percentage change in price for stock j. As the average holding period for a stock for short-term, medium-term, and long-term traders is 30-days, 200-days, and 580-days, respectively, we evaluate the 30-day expected RoR for short-term traders and 200-day RoR for medium-term traders and 580-day expected RoR for long-term traders.

We see from Table 4 below that users can expect to have positive returns if they follow our recommendations.

Table 5
95% Monte Carlo Simulation VaR Results

Segment	Average VaR ($)
Short Term (30 days)	163.35
Medium Term (200 days)	434.39
Long Term (580 days)	932.01

5.2. Value-at-risk

Value-at-Risk (VaR) is a popular financial metric used to estimate how much a portfolio may lose over a given period and confidence level (Linsmeier et al., 2000). In this study, we will be using Monte Carlo simulation to estimate the VaR of the recommended portfolios. The Monte Carlo simulation works by generating many trials with random market conditions for future returns and aggregates the change in value for each trial. VaR was defined by (Rockafellar & Uryasev, 2002) using equation 17 below:

$$VaR_\alpha(X) = min\{z \in \mathbb{R} : F_X(z) \geq \alpha\} \tag{17}$$

where α is the significance level and F_X(z) represents the cumulative distribution function of X with value z.

With an initial portfolio of $10,000, we performed 10,000 Monte Carlo simulations for 30 days for short-term traders, 200 days for medium-term traders, and 580 days for long-term traders and obtained their 95% confidence level. The results of the Monte Carlo simulation results are in Table 5 below.

From the results, we deduced that, on average, we are 95% confident that a short-term trader will not lose more than $165 in 30 days, a medium-term trader will not lose more than $435 in 200 days, and a long-term trader will not lose more than $933 in 580 days. These account for less than 2%, 5%, and 10% of the initial amount placed in the portfolio for the short-term, medium-term, and long-term segments, respectively. Therefore, we can conclude that our recommendations will not result in massive financial losses for the traders.

6. Discussion

6.1. Contributions to literature

We study the problem of building a stock recommender system that considers the needs and interests of the users. Previous research have noted the challenge of sparsity and scalability when using traditional CF methods (de Aguiar et al., 2020; Hayashi, & Takahashi, 2021; da Silva et al., 2018) that further affects the performance of the model. As such, in our study, to address the efficiency challenge in optimization, we first clustered the users based on their total spending on each stock before building a recommender system for the short-term, medium-term, and long-term traders. By doing so, we are not only effectively reducing the computational cost by scaling down the size of the dataset but also allowing the recommender system to have a more specific information on the users' interest, boosting the performance of our recommender system.

Moreover, as mentioned in Section 2, Prieto-Torres and Galpin (2020) proposed a non-personalized CF model to recommend stocks to users. However, providing personalized recommendations allow for better decision-making and enhance the user's overall experience, enabling the business to better retain its customer base and drive revenue (Basu, 2021; Tian et al., 2020). Therefore, in our study, we aimed to provide stock suggestions that cater to the user's trading strategy as well as their tastes and preferences.

While most recommender systems use explicit data as input, traders are rarely asked to rate the stocks that are available in the market. Furthermore, while multiple research exist on the usage of ARM as a recommender system (Wu et al., 2020; Osadchiy et al., 2019; Neysiani et al., 2019; Viktoratos et al., 2018), its application in the context of the stock market is limited. As such, in our study, we also compared the performance of using implicit data as input for our ARM model against explicit data as input for our other recommender systems. Our results suggests that recommender systems using explicit data outperformed those using implicit data – we noted that implicit data being innately noisy and a one-class problem with no negative feedback, in addition to our already highly sparse dataset, exacerbated the predictive efficacy of the recommender system and thus providing worse recommendations than the systems using explicit data.

Lastly, Ali et al. (2022) enhanced their decision support system by evaluating the VaR of the company in order to select which suppliers to work with. Expanding on this idea, we evaluated the VaR of the stocks that our recommender system has provided each user. By evaluating the financial risks that our recommendations may bring, we allow the traders to weigh their potential losses, thereby providing greater support in their risk management.

6.2. Implications for practice

The results of our extensive research have allowed our recommender system to become practical for large-scale data by first performing hierarchical clustering on the users and have shown that out of the three recommender systems that we have explored and evaluated, our kNN algorithm performed the best based on both predictive and classification accuracy metrics. Therefore, we can conclude that our kNN recommender system does not only predict the user ratings well but also provide relevant stock recommendations to the users.

In addition, as we acknowledge that users are wary of the financial losses that they may experience in the stock market, to support these users in making risk-informed decisions, we evaluated the Expected Rate of Return and Value-at-Risk of our recommended portfolios and showed that our recommendations are profitable.

6.3. Limitations and future research directions

As with most literature reviews, this research is subjected to several limitations. One of the main drawbacks of our study is that no information about the stocks, such as their stock symbols or company names, was available to us. Further, we also had no information on the trading experience of our app users and their risk tolerance. Information about the stocks the app users traded would have allowed us to better segment and group the stocks based on their market capitalization and sectors. The additional information about our app users and stocks traded would have deepened our understanding of our users' tastes and stock preferences, thereby improving our recommender system. For example, if a user only traded large-cap stocks, this would indicate that this user is more risk-averse and would not be interested in small-cap stocks. Therefore we could exclude these stocks when building a recommender system for those groups of users. However, the LDB dataset only provided the stocks' CUSIP numbers, which we used as our unique identifiers in our recommender system. CUSIP stands for Committee on Uniform Securities Identification Procedures and is a nine-character alphanumeric code used to identify U.S. and Canadian stocks. The American Bankers Association and the Standard and Poor's (Langager, 2021) manage the CUSIP codes and their respective stock symbols, and hence, there is a fee for the CUSIP codes.

With our aim of providing recommendations based on the users' tastes and preferences, obtaining users' transactional data was of paramount importance. However, due to privacy issues, such data is not easily available and therefore, although the LDB data is outdated, it was the best that we could obtain.

Moreover, our current model focuses on providing recommendations that cater to the needs and interests of the app users that have previously profited from their trades. We have based our recommender system on the assumption that app users who have profited from their trades will continue to profit as they have excellent stock trading expertise. We have demonstrated that these stock trading behaviors are still profitable, but it does not provide recommendations on when to buy and sell these recommended stocks.

Hence, in the future, we may incorporate the prediction of stock prices with our recommendations to include more users in the construction of our system and provide them with the knowledge of when they should buy and sell. We may also consider partnering with other trading platforms to obtain more recent and relevant data. Despite the limitations of our stock dataset, we have demonstrated that our recommendations are quantitative and systematic and are beneficial to the app users as their trades would not result in massive losses for them.

7. Conclusion

With the ever-growing volume of information available on the Internet, there has been greater demand for recommender systems to overcome the problem of information overload. Therefore, in this paper, we explored three different approaches that can be used to build a personalized stock recommender system to help users in their decision-making when investing in stocks. We first segmented the users into short-term, medium-term and long-term traders in recognition that different traders have different investment styles. We then performed hierarchical clustering within each of the segments to tackle the issue of scalability and applied collaborative filtering methods to recommend stocks to the users. We concluded that the k-Nearest Neighbour algorithm performed the best among the methods that we explored. From the results, we deduced that, on average, we are 95% confident that a short-term trader will not lose more than $165 in 30 days, a medium-term trader will not lose more than $435 in 200 days, and a long-term trader will not lose more than $933 in 580 days. These account for less than 2%, 5%, and 10% of the initial amount placed in the portfolio for the short-term, medium-term, and long-term segments, respectively. The expected rate of return for short term traders was 4.15%, for medium term traders 10.24% and for long term traders 23.17%. Finally, we showed that our recommendations are promising and do not bear massive risks by evaluating their Expected Returns and Value-at-Risks.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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